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# Install required libraries if needed (uncomment these lines in Colab)
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```
# !pip install textblob seaborn wordcloud
```

```
# Import libraries
```

```
import pandas as pd
```

```
from textblob import TextBlob
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from wordcloud import WordCloud
```

```
from google.colab import files
```

```
# Upload CSV file
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```
uploaded = files.upload()
```

```
filename = list(uploaded.keys())[0]
```

```
# Load dataset
```

```
df = pd.read_csv(filename)
```

```
# Show first few rows
```

```
print(df.head())
```

```
# Check for missing values
```

```
print(df.info())
```

```
# Drop missing values if any
```

```
df.dropna(subset=['text'], inplace=True)
```

```
# Sentiment Analysis using TextBlob
```

```
def analyze_sentiment(text):
    analysis = TextBlob(text)
    polarity = analysis.sentiment.polarity
    if polarity > 0.2:
        return 'Positive'
    elif polarity < -0.2:
        return 'Negative'
    else:
        return 'Neutral'

# Apply sentiment analysis
df['Sentiment'] = df['text'].apply(analyze_sentiment)

# Display sentiment counts
print(df['Sentiment'].value_counts())

# Plot sentiment distribution
sns.countplot(x='Sentiment', data=df, palette='coolwarm')
plt.title("Sentiment Distribution")
plt.xlabel("Sentiment")
plt.ylabel("Number of Conversations")
plt.show()

# Generate Word Cloud for each sentiment
for sentiment in ['Positive', 'Negative', 'Neutral']:
    text_data = ' '.join(df[df['Sentiment'] == sentiment]['text'])
    wordcloud = WordCloud(width=800, height=400,
        background_color='white').generate(text_data)
```

```
plt.figure(figsize=(10, 5))  
plt.imshow(wordcloud, interpolation='bilinear')  
plt.axis('off')  
plt.title(f'Word Cloud for {sentiment} Conversations')  
plt.show()
```

```
# Save processed CSV with sentiments  
output_file = "social_media_sentiment_results.csv"  
df.to_csv(output_file, index=False)  
files.download(output_file)
```